

Q & A about Haircutting Robots

From shuai.li@oulu.fi

Dear Student,

It is a great pleasure to supervise you for the project of this year. We kindly invite you to complete a short questionnaire by replying to this email with brief answers. The topic is robotic haircutting / haircutting robots (reference: <https://github.com/Dai0731csc/Robotic-Haircutting-Publications>).

1 Do you believe haircutting robots could be commercially launched within the next 10 years? Please explain why.

1. What questions do you have regarding AI support for robotic haircutting?
2. What questions do you have regarding the mechanical structure of haircutting robots?
3. What questions do you have regarding control and motion planning for robotic haircutting?
4. What are your thoughts on the market potential of robotic haircutting?
5. Do you have any other questions about this topic?

Thank you very much for your participation.

Best regards,

Steven

From shuai.li@oulu.fi

Dear All

Thank you for your effort in reading the questionnaire and giving me your answer, special thanks to those who responded my questions.

I am now sharing the information collected from you through the following shared editable link:

https://docs.google.com/document/d/1n9E_fTZDsDcJhpTz06M7lX74KFNMRF9D/edit?usp=sharing&ouid=101510555342718231595&rtpof=true&sd=true

For those who have responded, could you please add your detailed information in the contents (directly input in the shared file)

For those who have not yet responded, it is not late and you are welcome to put your thoughts directly in the file by following the same format.

For all, could you please check the questions given by others in the file, and give your thoughts if you have for any one that interests you the most (optional, if you do so, please do not forget to put your name aside so that we know it is you)

Overall, haircutting robotics is an emerging area. For human barbers, it is common to use scissors, combs, clippers for cutting. Actually, scissors can be replaced by electrical clippers in all situations and only combs and electrical clippers can do all the tasks. As to all electrical clippers we see on the market, it is inherently safe, which is guaranteed by the mechanical design (ask your barber next time when you go). Therefore, using a comb together with an electrical clipper can do all haircutting task safely.

For human barbers, what they do is to hold a comb using one arm while hold a clipper using another arm. With dual robotic arms, with one arm holding a comb while the other a clipper, it can do anything a barber can. The only thing left is to drive the robotic arm to different positions and orientations relative to heads, the same pose made by human barbers for the tasks of haircutting.

For human barbers, the haircutting skills are quite similar to human drivers.

The drivers using their eyes to see the traffic in front of the car, and the information collected by eyes is converted to decisions on whether turn the steering wheels left or right, whether turn the throttles high or low;

Similarly, for barbers, they use their eyes to see the "traffic" on the head (the hair situation of a person), and each step the barber makes a decision to steer the clipper in the hand (move the clippers up and down, left and right or tile to different directions), until the desired hairstyles is achieved.

When people design autonomous driving cars, they just put sensors on the car (e.g. , cameras), and passing the videos collected by the camera to a neural network for decision making and then the control decision is transmitted to steer the wheels, control the throttle. By mounting sensors, e.g., cameras on the clippers attached at the end tip of robotic arms, we can transmit the video collected from cameras to a decision making neural network and pass the decision to drive the motion of the robotic arms.

Those are end-to-end trained method for autonomous driving and for haircutting robots.

VLA (vision language action) is also becoming popular. In future, haircutting robots can also use it.

See technical details by following the publications by us at

(<https://github.com/Dai0731csc/Robotic-Haircutting-Publications>)

In addition, this is an open access hair simulator and feel free to test (<https://digital-salon.github.io/>)

Finally, we are the first group of people who are discussing the technical and fundamental problems of haircutting robots. It is my great pleasure to have you involved.

Feel free to shoot me emails for any discussions on this topic.

Steven

From harshchauhan457001@gmail.com

1. Do you believe haircutting robots could be commercially launched within the next 10 years? Please explain why.

I do not think haircutting robots can fully replace human barbers within the next 10 years. Haircutting is a very sensitive task because the robot operates extremely close to the human head, face, and neck, so safety becomes a major concern. Another challenge is that hair differs a lot in texture, density, and structure from person to person, which makes accurate computer vision and analysis difficult.

Unlike many other robotic tasks, mistakes in haircutting are not easily reversible. If the robot makes an incorrect cut, it remains visible for a long time, so the system requires extremely high precision and reliability.

However, I believe AI-assisted robotic haircutting systems could still become commercially useful for repetitive or standardized tasks under human supervision. In my opinion, the more realistic future is shared autonomy, where robots assist human barbers instead of completely replacing them.

2. What questions do you have regarding AI support for robotic haircutting?

I am particularly interested in how AI and computer vision systems can accurately understand different hair types, textures, densities, and structures in real time. Since hair behaves differently for every person, perception and segmentation become very challenging problems for robotic haircutting systems.

I am also curious about how learning models would be trained for such tasks, especially because collecting large-scale haircutting datasets seems difficult. Another question I have is how these systems can adapt dynamically if the user slightly moves their head or if the hair behaves differently during the cutting process.

Personalization is another area that interests me a lot. Hair and hairstyles are very personal to individuals, so I believe robotic haircutting systems can only become truly useful if AI is able to understand user preferences and provide personalized results instead of only standardized cuts.

3. What questions do you have regarding the mechanical structure of haircutting robots?

One of my biggest questions regarding the mechanical structure of haircutting robots is how safety mechanisms are implemented, since the robot operates very close to sensitive areas such as the face, ears, and neck. Even a small mechanical or control error could lead to

serious consequences, so I am interested in how researchers design fail-safe systems, emergency stopping mechanisms, and safe human-robot interaction.

I am also interested in how robotic systems can achieve the level of precision required for high-quality haircuts. Maintaining stable and accurate tool movement while dealing with different hair structures and possible user movement seems very challenging. If robotic haircutting systems are able to consistently improve precision and repeatability, I think it could significantly improve the quality of results and increase the market potential of such systems.

4. What questions do you have regarding control and motion planning for robotic haircutting?

I am interested in how control and motion planning systems adapt when the customer slightly moves their head during the haircutting process. Since haircutting requires high precision near sensitive areas, real-time adjustment and safe replanning seem very important.

I am also curious about how haircutting trajectories are generated for different hairstyles and hair structures. It would be interesting to understand whether these paths are manually designed, generated similarly to CNC toolpaths, or learned through AI-based approaches.

5. What are your thoughts on the market potential of robotic haircutting?

I do not think there is a strong market for fully robotic haircutting yet, mainly because users still have concerns regarding safety and trust. However, I think the field could create a strong market position in the future if it is able to solve problems related to precision and consistency.

One advantage robotic systems may provide over human barbers is highly consistent and repeatable results according to a person's hair structure and head shape. Building user trust will still be a major challenge, especially regarding safety, but I think adoption could increase if users begin to see practical advantages and better consistency in their haircutting experience.

From ankitpal862005@gmail.com

Q-1. Do you believe haircutting robots could be commercially launched within the next 10 years? Please explain why.

Yes, I believe haircutting robots could be commercially launched within the next 10 years. In fact, academic and industry analyses within the provided literature strongly support a timeline significantly shorter than a decade, projecting initial commercial applications to emerge within the next 5 years (by roughly 2030).

1. The Supervised Autonomy Business Model

Achieving full, un-supervised robot autonomy for every single hair type and human edge-case is a massive hurdle that remains currently unfeasible. However, the industry does not require full autonomy to launch commercially. By adopting a model of supervised autonomy (much like partial autonomy in self-driving cars), a single human barber can oversee a localized fleet of multiple robotic units (e.g., managing five units simultaneously). The robot handles the repetitive, precision-oriented bulk or coarse trimming, and the human supervisor only

intervenes during unexpected events, motion faults, or specialized finishing requests. This dramatically reduces immediate labor costs and sidesteps the immediate need for "perfect" AI, creating a practical path to early commercialization.

2. Rapidly Decreasing Hardware Costs

Historically, complex humanoid control systems and high-precision multi-axis manipulators were financially restrictive. For instance, Honda's ASIMO robot cost roughly USD 1.3 million two decades ago. Today, highly capable humanoid platforms and collaborative robotic arms are available at a mere fraction of that cost—such as the Unitree R1 retailing for approximately USD 5900. This democratization of mechatronics and optimized global supply chains make setting up a robotic salon economically viable for businesses for the first time.

3. Clear Technology Transfer from Mature Fields

Haircutting automation is not starting from scratch; it heavily borrows proven algorithmic and mechanical frameworks from mature industries:

- **Precision Manufacturing:** The conceptualization of haircutting as a 5-axis simultaneous Computer Numerical Control (CNC) process utilizes mature toolpath generation, surface contouring, and multi-pass finishing strategies from industrial manufacturing.
- **Service & Household Robotics:** Global coverage strategies build directly on decomposition and mapping paradigms refined by household vacuum cleaners (like the iRobot Roomba) and autonomous lawnmowers.
- **Medical & Surgical Robotics:** High-precision, safety-critical close-contact control mechanisms—including real-time force feedback, sub-millimeter positioning, and variable impedance control—leverage standard frameworks established by systems like the da Vinci surgical platform.

4. Integration of Generative AI and Multimodal Sensing

The explosion of advanced foundational models provides the exact cognitive layer needed for high-quality human interaction. Integrating Large Language Models (LLMs) allows the robot to seamlessly parse casual natural language inputs (e.g., transforming a requested "fade cut" into localized depth adjustments and specific clipper-guard paths). Simultaneously, multi-sensor fusion configurations combining vision, depth cameras, high-frequency IMUs for head tracking, and acoustic tracking allow the system to continuously map and safely adapt to involuntary head movements in real time.

Q-2 What questions do you have regarding AI support for robotic haircutting?

1. Handling Different Hair Types

Right now, the computers inside these robots are mostly trained on straight or wavy hair. But what happens when a person with really thick, curly, or coily hair sits in the chair? Curly hair stretches when you pull it and shrinks back up when you let go. How can the AI predict exactly how long the hair will actually be after it's cut?

2. Dealing with People Moving

A haircut takes a long time, and people naturally wiggle, tilt their heads, or blink. If the robot's camera sees the customer move, how does the AI decide whether to just slightly nudge the scissors over in real-time, or completely stop everything, recalculate the entire path, and start over? How does it make sure it doesn't jam the clipper into a thick patch of hair too fast and pull on the customer's scalp?

3. Measuring a "Good" Haircut

In regular factory manufacturing, a robot knows it did a perfect job if a metal part is accurate down to a millimeter. But a haircut is art, not just math. How can we teach an AI to look at a finished haircut and actually tell if it looks "good," symmetric, nicely blended, and stylish, rather than just checking if the lengths match a rigid blueprint?

4. Stopping Mistakes Before They Happen

Some engineers want to use the latest "smart" AI models so the robot can just listen to a voice command like "give me a clean fade" and immediately start moving the arm. But AI is notorious for guessing or making things up sometimes. Since a human head is centimeters away from a sharp moving blade, how do we build a digital "safety cage" around the AI's brain so that even if it misunderstands a word, it physically cannot move the blade into a danger zone?

5. Learning from Human Barbers

The business plan for these salons is to have one human master barber watching over five robots at the same time. If a robot gets confused by a weird head shape and the human has to grab the controls to finish a tricky spot, how does the robot learn from what the human just did? How does the AI absorb that quick lesson without accidentally "forgetting" its core safety rules?

Q-3 What questions do you have regarding the mechanical structure of haircutting robots?

1. The Setup

- Which physical setup makes the most sense for a real shop: a giant box frame that surrounds your head but stays incredibly steady, or a single robot arm that looks friendlier but might wobble?

2. Finding the Balance

- How do you make a robot gentle enough not to bruise or cut you if you sneeze, but stiff enough that it doesn't just bend backward when trying to slice through a thick chunk of hair?

3. Handling the Prep Work

- How do you physically build a robot hand that can delicately comb out a knot or clip hair into neat sections without pulling it out by the roots?

- Does the machine need a single multi-tool hand like a Swiss Army knife, or does it have to constantly stop the haircut to swap tools?

4. What Happens if the Power Fails?

- What are the physical, built-in safety nets that have absolutely nothing to do with computers or code to keep a heavy metal arm from collapsing onto your neck if the power suddenly cuts out?
- Are there mechanical springs or manual brakes that instantly snap the blades away from your skin the second the electricity drops?

5. Surviving the Mess

- How do you design a heated styling tool that won't accidentally burn a customer's scalp if the arm pauses for too long?
- How do you seal the robot's moving joints so that thousands of tiny, prickly hair clippings don't get inside the gears and ruin the expensive machine after a few weeks of work?

Q-4 What questions do you have regarding control and motion planning for robotic haircutting?

1. Moving Around a Moving Target

- How does the robot smooth out its movements so it doesn't give someone a choppy, uneven haircut while trying to keep up with a head that is naturally tilting, nodding, or shifting in the chair?
- If a customer makes a sudden, fast movement—like a sharp jerk from a cough or a sneeze—how does the robot instantly alter its path to dodge their ears or eyes without its mechanical arm locking up or lagging?

2. Managing the Speed and Pressure

- How does the robot know how fast to move the clippers through the hair?
- How does it tell the difference between hitting a thick patch of dense hair that just needs a slower push, and actually pressing too hard against the customer's actual scalp?

3. Cutting in "Layers" or Steps

- Since a robot can't cut all the hair down to the right length in just one single pass, how does it plan out the steps for a multi-pass cut?
- How does it remember exactly where it has already trimmed and where it still needs to go, especially after the customer moves their head around mid-haircut?

4. Splitting Up the Work

- How does the robot mathematically map out and break down a human scalp into different workspace sections (like the crown, the sides, and the back of the neck) to tackle them one by one?
- Once it splits the head into those sections, how does it smoothly blend the boundaries between them so the customer doesn't end up with harsh, weird lines?

5. Transitioning from Human to Robot Control

- If the robot gets confused by a complicated hairstyle and a human barber has to take over the controls remotely, how does the robot smoothly hand over the reins without a sudden, jerky movement that could nick the customer?
- While the human supervisor is guiding it, how does the robot's internal mapping system still keep up its automated safety boundaries in the background to prevent human operator error?

Q-5 What are your thoughts on the market potential of robotic haircutting?

1. A Massive, Untapped Global Footprint

The global hairstyling industry is an enormous economic market, currently exceeding USD 50 billion annually. Right now, it is almost entirely manual. When you break down the global population into spending tiers, the math shows a massive ceiling for automated services:

- High-income regions alone contribute an estimated USD 576 billion annually to the overall beauty and personal care market space, driven by a high average monthly per capita spending of around USD 40.
- Middle-income and low-income regions combine for roughly another USD 250 billion annually.
- Major metropolitan areas highlight this highly localized wealth density; for example, male haircut expenditures alone account for approximately USD 3.96 billion in New York City, GBP 3.49 billion in London, and RMB 4.5 billion in Shanghai every single year.

Capturing even a tiny sliver of this daily routine service through automation represents a brand-new, highly lucrative sector in the service robotics industry.

2. Powerful Macro Drivers: Labor and Demographics

The demand for these machines isn't just about cool technology; it's being forced by real-world economic pressures:

- Labor Shortages & Costs: The rising cost of skilled human labor and severe workforce shortages in personal care services are making automated alternatives highly in-demand for shop owners who want to keep margins viable.
- Aging Populations: As the global population ages, the need for accessible, automated personal grooming in assisted living facilities and healthcare environments is scaling up rapidly.
- The Post-Pandemic Shift: The COVID-19 pandemic permanently altered public awareness around hygiene, driving sustained commercial interest in contactless, highly consistent personal care options.

3. The Financial Breakthrough: Democratized Hardware

For decades, building a haircutting robot was economically impossible because the mechanical parts were too expensive. Two decades ago, a highly advanced robot like Honda's ASIMO cost around USD 1.3 million.

Today, global supply chain optimizations and mechatronic advancements have crashed those barriers. With capable, modern collaborative components and humanoid systems (like Unitree's R1) hitting the market for an estimated retail price of just USD 5,900, the upfront cost to build and install a robotic workstation is low enough to yield a rapid return on investment for small business owners.

4. A Realistic Roadmap via Supervised Autonomy

We don't need "perfect AI" that can handle 100% of haircuts autonomously for this market to explode. By mimicking the autonomous driving industry and using a supervised autonomy business model, early adoption is incredibly practical.

Instead of a traditional one-barber-to-one-customer salon layout, a single human stylist can oversee five robotic units at once. The robots efficiently handle the repetitive, precise, and time-consuming bulk cutting, while the human steps in only for tricky detail passes or customized styling adjustments. This scales up salon throughput, slashes consumer wait times, and drastically lowers overhead costs.

5. High-Potential Early Market Segments

Because style expectations vary and safety regulations take time to clear, the rollouts won't happen everywhere at once. The best immediate market opportunities lie in high-volume, highly standardized environments:

- **Budget Quick-Trim Salons:** Located in urban centers, shopping malls, airports, or corporate campuses, providing fast, affordable, and highly consistent maintenance trims.
- **Institutional Environments:** Military bases (where strict, uniform haircuts are mandatory) and remote operational camps.
- **Assisted Care Facilities:** Providing hygienic, dignified, and easily accessible grooming directly to individuals with disabilities or limited mobility right in their living spaces.

Q-6 Do you have any other questions about this topic?

1. The Vibe and the Conversation

- Barbershops and salons are huge social hubs where people go to chat, vent, and connect with their community. If you replace the stylist with a machine, do we lose that entire comforting social ritual? Even if the robot has a built-in AI voice to make small talk, would it just feel completely hollow and transactional compared to a real human connection?

2. The Comfort and Anxiety Factors

- Having sharp, fast-moving metal blades and high-heat tools operating millimeters away from your eyes and neck requires a massive amount of trust. How will businesses overcome the

natural "fear factor" of sitting in that chair? Will customers spend the entire hour tensed up and stressed out, completely ruining the relaxing, pampering experience that people usually look forward to at a salon?

3. Handling Mistakes and "Bad Haircut" Regret

- If a human stylist messes up your layers, you can talk it through, express your disappointment, and they will do their best to adapt, blend it out, or offer a discount to make it right. But what happens if the robot suffers a minor software glitch and accidentally buzzes a random patch down to the skin? Who takes the blame, how does a machine handle an angry customer, and what is the protocol for fixing a physical mistake that takes months to grow back?

4. Pricing and the Accidental "Luxury" Divide

- If these robot stations cost around USD 5,900 and allow one stylist to run five booths at once, basic haircuts should theoretically become incredibly cheap and fast. But does this mean getting a haircut from a real, living human will turn into an expensive luxury that only wealthy people can afford? Will everyday people be priced out of human touch and forced to use automated "haircut vending machines" for basic personal care?

5. Keeping Up with Fast-Moving Trends

- Human barbers stay relevant by watching social media, seeing what celebrities are doing, and instantly figuring out how to replicate a cool new messy texture or a modern trend on the fly. If a customer walks in and says, "I want that exact style that went viral on TikTok this morning," how long does it take for the AI to model it, test it for safety, and program the robot to execute it? Will robots always be a step behind human creativity?

From upendrasingh6537@gmail.com

Dear Dr. Li,

Thank you very much for your email and for the opportunity to participate in this project. Please find my brief responses below.

I believe robotic haircutting could become commercially viable within the next 10 years, especially for standardized or assistive haircutting tasks. Advances in computer vision, force sensing, dexterous robotic manipulation, and generative AI are progressing rapidly. However, full commercial deployment will depend heavily on safety, reliability, user trust, and the robot's ability to adapt to different hair textures, styles, and unpredictable human movement.

Regarding AI support for robotic haircutting, I am interested in:

- How AI can accurately perceive and model different hairstyles and hair properties in real time.
- Whether multimodal AI systems can understand customer preferences from text, images, or voice instructions.

- How reinforcement learning or imitation learning could be applied safely in human-interactive haircutting tasks.
 - How AI can ensure consistency while still allowing personalized styling.
3. Regarding the mechanical structure of haircutting robots, I would like to know:
- What type of robotic manipulator design is most suitable for safe close-contact interaction with humans.
 - How compliant or soft robotic mechanisms could improve safety and adaptability.
 - How end-effectors are designed for handling scissors, clippers, and combs.
 - What sensing systems are integrated for collision avoidance and force feedback.
4. Regarding control and motion planning, I am curious about:
- How real-time trajectory planning is achieved while accounting for head motion and hair deformation.
 - How safety constraints are incorporated into motion control near sensitive regions such as the face and ears.
 - Whether shared autonomy between human stylists and robots is considered.
 - How uncertainty in perception and manipulation is handled during cutting operations.
5. I think the market potential is significant, particularly in:
- High-volume salons seeking automation assistance.
 - Elderly care or assistive grooming applications.
 - Regions with labor shortages.
 - Personalized AI-driven styling services.

At the same time, customer acceptance and trust will likely determine adoption speed. Initially, hybrid human-robot collaboration models may be more commercially successful than fully autonomous systems.

6. One additional question I have is how ethical, legal, and liability issues would be handled if a robotic haircutting system causes injury or unsatisfactory results. I am also interested in understanding what level of autonomy researchers envision for future systems.

Thank you again for organizing this questionnaire. I look forward to learning more about this research area.

Best regards,

Upendra Singh

IIT Kharagpur

From maitrig69@gmail.com

Respected Professor Li,

Thank you very much for the opportunity to participate in this project. Please find my responses below.

1. Do you believe haircutting robots could be commercially launched within the next 10 years? Please explain why.

Yes, I believe haircutting robots could achieve commercial deployment within the next 10 years, especially in supervised or semi-autonomous settings. Recent advances in AI, computer vision and humanoid robotics have significantly improved the feasibility of safe human-robot interaction near sensitive regions such as the head and face. However, I think the first large-scale deployments will likely involve shared autonomy or teleoperation with human supervision rather than fully autonomous systems.

2. What questions do you have regarding AI support for robotic haircutting?

- How can AI reliably model and track deformable hair structures in real time across different hair textures, densities, and curl patterns?
- What datasets would be required for large-scale training of haircutting robots, especially considering privacy concerns with facial and scalp data?
- How can AI evaluate haircut quality objectively, considering that aesthetics are subjective and culturally dependent?

3. What questions do you have regarding the mechanical structure of haircutting robots?

- What type of robotic configuration would be most suitable: humanoid arms, collaborative robotic arms, or specialized fixed-structure systems?
- What trade-offs exist between lightweight designs and mechanical rigidity required for precise cutting?
- How should end-effectors be designed to safely integrate clippers, scissors, dryers, or combs while minimizing accidental injury?
- Would multi-arm systems provide practical advantages for simultaneous combing, sensing, and cutting?

4. What questions do you have regarding control and motion planning for robotic haircutting?

- How can real-time head pose estimation and motion compensation be integrated with trajectory planning to handle involuntary human movement?
- What types of safety-aware control algorithms are most effective near sensitive anatomical regions such as ears and eyes?

5. What are your thoughts on the market potential of robotic haircutting?

I think the market potential is very large. Haircare is a recurring global service industry with extremely high cumulative spending. Robotic haircutting could reduce labor dependency, improve accessibility, and enable scalable grooming services in locations such as malls, airports, hospitals, and remote regions. Initially, adoption may occur in standardized haircut styles or assisted-service environments, but over time the technology could expand into personalized grooming and styling ecosystems supported by AI and digital platforms.

Thank you again for the opportunity. I am excited to learn more about this interdisciplinary area.

Best regards,

Maitri Gupta

From rahul7541kumar1@gmail.com

Thank you for the warm welcome. Please find my responses to the questionnaire below:

1. Do you believe haircutting robots could be commercially launched within the next 10 years?

Yes, I believe it is feasible. With rapid advancements in robotics, computer vision, and AI, the core technical challenges are becoming increasingly solvable. However, commercial success will also depend on cost reduction, safety certification, and public acceptance — all of which are achievable within a decade with focused research.

2. What questions do you have regarding AI support for robotic haircutting?

How can vision-language-action (VLA) models be made robust enough to handle the high variability in hair types, head shapes, and hairstyle preferences? What is the best approach to train such models with limited real-world data?

3. What questions do you have regarding the mechanical structure of haircutting robots?

What end-effector design is most suitable for safely handling scissors or clippers near a human head? How can the robot maintain stable contact force on a deformable and moving surface?

4. What questions do you have regarding control and motion planning for robotic haircutting?

How can the robot adapt its planned paths in real time if the person moves their head? What control strategies best balance speed and precision during cutting?

5. What are your thoughts on the market potential of robotic haircutting?

The global haircare market is large and the demand for affordable, consistent haircuts is high. Robotic haircutting could be very attractive for high-volume, low-cost salons, elderly care facilities, and regions with limited skilled labor. The market potential is significant if safety and quality can be guaranteed.

6. Do you have any other questions about this topic?

How do you envision the human-robot interaction during a haircut — will the user be able to provide real-time feedback or corrections to the robot mid-process?

Thank you for this opportunity. I look forward to contributing to this project.

Best regards,

Rahul Kumar

Department of Mining Engineering (AI/ML Micro-Specialization)

IIT Kharagpur

rahul7541kumar1@gmail.com | +91-7737408932

From Soham Padsumbia<sohampadsumbia2006@gmail.com

1. I believe that with the progress so far and the sentiments of people, In India haircutting robots will be in their phase 1 of market entry after 10 years, it will take a lot of marketing and awareness among people to accept that they could trust a robot for their haircut, on the other hand, the idea of cutting hair via robot sounds cool and innovative, so we need to leverage this and attract as many young customers as possible. I believe that the quality of the product is also important here, if there is even a single complaint, then this whole concept may be scrapped.

2. AI however advanced it might be, how will it be able to identify and cater the customized needs of the consumer, how will the customer express the type of haircut he or she wants?

3. Will the structure be human like or will they look like robots? also since robots will be made by hard material how will you ensure that they dont hurt the person while haircutting? Also the touch of the robot should be smooth, otherwise the experience of the customer will be unpleasant and we wont get good retention.

4. I still am figuring out the mechanical part and hence dont have questions till now.

5. I believe that the market for robotic arm would be new-generational youth, and we should market it as a innovative one-time experience at

the start. Still the addressable market is pretty big and hence I think it might fit in.

from malakarg95@gmail.com

Dear Professor Li,

Thank you for the warm welcome and for sharing the reference. I have gone through the publications and am delighted to be working on this project with you. Please find my responses below.

1. Do you believe haircutting robots could be commercially launched within the next 10 years?

Yes, I do. The convergence of high-precision robotic arms, real-time 3D sensing, and recent advances in Vision-Language-Action models has brought this goal within reach. The key remaining challenges — handling head motion, varied hair textures, and safety guarantees — are active research problems with tractable solutions. I expect commercial pilots in controlled salon environments (targeted at elderly care or individuals with mobility limitations) within the decade, with broader deployment following maturation of sim-to-real transfer methods.

2. What questions do I have regarding AI support for robotic haircutting?

I am most curious about how VLA models can be fine-tuned with limited real-world demonstration data, given the cost of collecting paired hairstyle-action sequences. Specifically: what is the minimum dataset size needed for reliable generalisation across hair types and styles, and how should uncertainty be communicated back to the robot's control loop when the model encounters an out-of-distribution input?

3. What questions do I have regarding the mechanical structure?

I wonder about the trade-off between end-effector precision and compliance — a rigid arm offers repeatability but may be unsafe upon unexpected head movement, while a compliant design absorbs disturbances but sacrifices cutting accuracy. Additionally, I am curious about tool design: whether a single multi-function end-effector (trimming, clipping, shaving) is viable or whether a modular tool-change approach is preferred for commercial systems.

4. What questions do I have regarding control and motion planning?

My primary question concerns real-time re-planning: how frequently must the path planner update when the head moves, and what latency budget is acceptable before cutting quality degrades? I am also interested in how surface-normal-aligned tool orientation is maintained during curved-surface traversal, and how path planning integrates safety exclusion zones (ears, eyes, neckline) derived from perception.

5. Thoughts on market potential?

The market potential is substantial, particularly across three segments: (i) elderly and disability care, where independent grooming is a quality-of-life issue, (ii) high-volume commercial salons seeking cost and speed optimisation, and (iii) remote or underserved regions lacking

skilled barbers. Global hairdressing services represent a market exceeding USD 200 billion annually. If a robot system can deliver consistent, safe results at an accessible price point, the addressable opportunity is significant.

6. Any other questions?

I am keen to understand how the research community currently evaluates robotic haircutting systems — whether standardised benchmarks exist for cutting accuracy, safety events, and style fidelity — and how the reference publications in the GitHub repository approach ground-truth measurement. I would also appreciate guidance on which aspects of this project you consider highest priority for our collaboration this year.

Thank you again for the opportunity. I look forward to your guidance.

Best regards,

Saugata Malakar

3rd Year Undergraduate

Department of Civil Engineering,

Indian Institute of Technology, Kharagpur

+91 8240541867

malakarg95@gmail.com

from jagrutpatel999@gmail.com

Dear Professor,

Thank you for the opportunity. I am very excited to work with you on this project.

Here are my answers to the questionnaire:

1. I think it is very likely we will see a commercial launch for basic haircuts within the next 10 years. While complex, highly styled cuts might take longer, standard trims or buzz cuts are manageable. The main challenge isn't just the cutting itself, but safety and building consumer trust. As sensor technology and real-time AI processing get faster, those safety barriers will become much easier to overcome.

2. How will the AI handle unexpected cases, like a person suddenly moving their head and I am also curious about how we might use multimodal AI like could the system take a reference photo and voice instructions from the user and translate that directly into cutting parameters?

3. how many degrees of freedom does the robotic arm need to comfortably and smoothly reach every angle of a human head without being too bulky?

4. I have spent time working on 6-DoF motion forecasting, so I am very interested in the real-time latency aspect of this. How do we ensure the motion planning algorithms react instantly to micro-movements?

5. The market potential is huge, particularly in areas with high labor costs or for high-volume, standardized cuts. Beyond regular salons, I could see a strong market for this in hospitals, care homes, or military bases where quick, hygienic, and efficient haircuts are needed.

6. Are there specific simulation environments or datasets we will be using to test our models safely before deploying anything to physical hardware?

From dhruvgupta11yo@gmail.com

Name: Dhruv Gupta

College: Indian Institute of Technology Kharagpur

Branch- Metallurgical and Materials Engineering with Micro specialization in Artificial Intelligence

Exploring the domain of Machine Learning and its application in Material Science, particularly in Battery electrochemistry and Use of Machine learning and data science for Material discovery.

Hello Sir,

I hope you are doing well. These are my thoughts on the questions.

Question 1: Yes, but only for simple cuts. We will probably see them in shops within 10 years, but they will only do basic things like simple trims or buzz cuts. The physical robots are already smart enough, but making them 100% safe and getting people to actually trust a machine with blades near their face will take some time.

Question 2:

1. How does the AI scan a head properly if the person has thick, curly hair that covers their ears?
2. If someone sneezes or moves suddenly, how fast can the AI tell the robot to stop cutting?
3. Can the AI look at a picture of a haircut I want and actually figure out how to make it look good on my specific head shape?

Question 3:

1. Will the robot have two arms to mimic how a human uses one hand for scissors and one for a comb?
2. What happens if the robot gets stuck or loses power? Is there a backup system that instantly pulls the blades away?

3. Hair gets everywhere. How do you keep tiny hair clippings from getting inside the robot's joints and breaking it?

Question 4:

1. Hair moves and bends when you touch it. How does the robot plan its movements when the hair keeps changing shape?
2. Can the robot feel if it is pulling someone's hair or pushing too hard against their skin?
3. How do you program it to make absolutely sure it never goes near sensitive areas like the eyes or ears?

Question 5: I think there is a big market for quick, cheap, and basic haircuts, like at military bases, retirement homes or budget walk-in shops. But for expensive salons, it will be hard. Most people go to a barber for the conversation and the relaxing experience, which a robot can't really replace.

Question 6:

1. If a robot accidentally cuts someone, who gets in trouble? The shop owner, or the company that built the robot?
2. These robots will probably cost a lot of money. Will they actually save salon owners money compared to just hiring human barbers?

These were my answers to the questions.

Best Regards,

Dhruv Gupta

From vishnuvamsi7482@gmail.com

Dear Professor,

Thank you for the opportunity to work on this project under your supervision. Please find my responses to the questionnaire regarding robotic haircutting below:

1. Do you believe haircutting robots could be commercially launched within the next 10 years ?

Yes, due to rapid advancements in precision robotics and AI safety protocols.

2. What questions do you have regarding AI support for robotic haircutting ?

How can computer vision effectively handle diverse hair textures and head shapes ?

3. What questions do you have regarding the mechanical structure of haircutting robots ?

What safety mechanisms are prioritized to prevent accidental contact with the skin ?

4. What questions do you have regarding control and motion planning for robotic haircutting ?

How is the real-time path planning adjusted for small movements made by the user ?

5. What are your thoughts on the market potential of robotic haircutting ?

I believe there is significant potential in automated salons and assisted living facilities. However, in developing economies, traditional haircutting methods may still be prioritized due to cost-effectiveness.

6. Do you have any other questions about this topic ?

I have no further questions at this time.

I look forward to our collaboration on this project.

Best regards,

Name : Vishnu Vamsi

Roll No : 22EC39020

From aibcs2022@gmail.com

Dear Professor,

Thank you for giving me the opportunity to contribute to this project under your guidance. Please find below my responses to the questionnaire on robotic haircutting.

1. Do you believe haircutting robots could be commercially launched within the next 10 years? Please explain why.

Yes, I believe commercially viable haircutting robots can emerge within the next 10 years, at least for constrained use cases. Initially, they may focus on standardized hairstyles such as buzz cuts, fades, trimming, or maintenance cuts before handling highly artistic or complex hairstyles.

Several technologies are already progressing rapidly:

- robotic manipulation,
- real-time computer vision,
- force feedback systems,
- imitation learning,
- VLA (Vision-Language-Action) models,
- human pose and head tracking.

The main challenge may not be the cutting itself, but achieving the reliability, safety, adaptability, and customer trust required for real-world deployment. Human hair is highly deformable and varies significantly in texture, density, curl, and motion, which makes precision difficult.

I think hybrid systems may appear first, where a human supervises or assists the robot, similar to semi-autonomous driving systems.

2. What questions do you have regarding AI support for robotic haircutting?
 - How do you plan to represent hair geometry digitally for AI decision making?
 - Will the AI rely mainly on RGB vision, depth sensing, or multimodal sensing?
 - How will the system handle occlusions caused by dense hair or comb positioning?
 - Are you considering imitation learning from expert barbers?
 - How will the AI evaluate haircut quality during intermediate stages rather than only final output?
 - Can reinforcement learning realistically be used given the safety constraints around human heads?
 - How much hairstyle generalization do you expect from one trained model?
 - Could language-conditioned models eventually allow commands like “medium fade with textured top”?
3. What questions do you have regarding the mechanical structure of haircutting robots?
 - What type of robotic arm configuration is being considered?
 - How many degrees of freedom are necessary for human-level flexibility?
 - Will compliance or soft robotics be incorporated for safety?
 - How will the robot maintain stable relative positioning if the customer moves slightly?
 - Is force sensing integrated near the end effector?
 - How will the system handle tool changing between clipper, comb, dryer, or other tools?
 - Are there plans for lightweight robotic arms optimized specifically for salon environments instead of industrial robotic arms?
4. What questions do you have regarding control and motion planning for robotic haircutting?
 - Will the motion planning be trajectory-based or fully reactive in real time?
 - How will collision avoidance near ears, eyes, and scalp be guaranteed?
 - How frequently is replanning performed during cutting?
 - Is there a digital head model maintained during operation?
 - How do you plan to synchronize comb positioning and clipper motion across dual arms?
 - Will the robot imitate demonstrated human barber trajectories or learn optimized trajectories independently?
 - How will uncertainty estimation and confidence scoring be handled during motion execution?
5. What are your thoughts on the market potential of robotic haircutting?

I think the market potential is significant in the long term, especially in:

- high-volume salons,
- elderly care,

- military grooming,
- hospitals,
- airports,
- smart retail environments,
- regions with labor shortages.

A robotic system capable of delivering fast, repeatable, low-cost haircuts could become commercially attractive. However, customer acceptance may initially depend on trust, perceived safety, hygiene, and haircut consistency.

I also think personalization will become important. Customers may eventually store digital hairstyle profiles and reproduce them consistently across locations.

6. Do you have any other questions about this topic?
 - How do you plan to benchmark haircut quality objectively?
 - Are there public datasets for robotic haircutting research?
 - What safety certification challenges do you anticipate?
 - Do you believe fully autonomous haircutting will surpass human barbers in consistency for standardized styles?
 - Could simulation environments realistically model hair physics sufficiently for large-scale training?
 - How important do you think tactile sensing will become compared to pure vision-based approaches?

Thank you again for sharing this research direction. I think it is a very interesting interdisciplinary area involving robotics, AI, computer vision, human-robot interaction, and control systems.

Best regards,

Rohan

from dr.asislaha.cil@gmail.com

Respected Professor Li,

Thank you for the opportunity to be a part of this project. I found the topic really interesting, and my responses are given below.

Q1. Do you believe haircutting robots could be commercially launched within the next 10 years? Please explain why.

Yes, I think haircutting robots could become commercially available within the next 10 years, at least for simpler or supervised tasks. AI, robotics, and computer vision have improved a lot in recent years, especially in areas involving precise movement and human interaction. I feel

the first versions will probably work alongside human supervision rather than being completely autonomous.

Q2. What questions do you have regarding AI support for robotic haircutting?

- How can AI handle the large variation in hair textures, lengths, and styling preferences between different people?
- What would be the safest way to collect training data for such systems while protecting user privacy?
- Can AI systems understand whether a haircut “looks good,” considering that style preferences are very subjective?

Q3. What questions do you have regarding the mechanical structure of haircutting robots?

- What type of robotic structure would work best for haircutting while still being safe and comfortable for users?
- How can tools like clippers or scissors be designed to reduce the risk of accidental cuts near the face or ears?
- Would using softer or more flexible robotic components make the system safer for close human interaction?

Q4. What questions do you have regarding control and motion planning for robotic haircutting?

- How can the robot react safely if the person suddenly moves during the haircut?
- What kind of control system would help maintain both accuracy and smooth movement?
- How can the robot maintain gentle contact with the head without causing discomfort?

Q5. What are your thoughts on the market potential of robotic haircutting?

I think the market potential is quite strong, especially in places where people value speed, consistency, and convenience. Such systems could be useful in busy urban areas, shopping malls, airports, or even elderly care facilities. At first, they may mainly focus on simpler hairstyles, but over time they could expand into more personalized grooming services.

Q6. Do you have any other questions about this topic?

How important do you think customer trust and comfort will be for the success of robotic haircutting systems? Even if the technology becomes highly accurate, I feel people may still hesitate initially about allowing a robot to perform such a close-contact task.

Best Regards,

Roshni Laha

2nd year undergraduate

Department of Electrical Engineering

IIT Kharagpur

from souptik8@gmail.com

Dear Professor,

Thank you for sharing this interesting research topic. After reading the provided materials, I have noted down some of my thoughts and questions below.

1. Do you believe haircutting robots could be commercially launched within the next 10 years? Please explain why.

Yes, I think commercial deployment is possible within the next decade, particularly for supervised or semi-autonomous systems. Recent progress in robotic perception, force control, and multimodal AI seems to make close-contact grooming tasks increasingly feasible. However, I think fully autonomous haircutting will still require major advances in safety assurance, real-time adaptation to human movement, and aesthetic decision-making.

2. What questions do you have regarding AI support for robotic haircutting?

- How can AI models robustly perceive hair geometry when hair properties vary significantly across individuals?
- Could vision-language-action systems learn personalized haircut preferences through interaction with users?
- To what extent can simulation environments replace risky real-world data collection for training?

3. What questions do you have regarding the mechanical structure of haircutting robots?

- What mechanical design would best support safe operation near sensitive regions such as the scalp, ears, and neck?
- Would lightweight or compliant robotic structures improve both safety and user comfort?
- How might future systems integrate multiple grooming tools without making the hardware overly complex?

4. What questions do you have regarding control and motion planning for robotic haircutting?

- How can motion planners maintain both precision and safety when the customer moves unexpectedly?
- What role could force sensing and tactile feedback play during close-contact cutting operations?

- Would hybrid approaches combining classical robotics and AI-based control be more reliable than fully end-to-end learning systems?

5. What are your thoughts on the market potential of robotic haircutting?

I think the market potential is significant, especially in high-density urban areas and regions facing labor shortages in service industries. Initially, robotic haircutting may function more as an assistive technology for salons rather than a complete replacement for human barbers. Over time, improvements in affordability, reliability, and customer trust could expand adoption further.

6. Do you have any other questions about this topic?

I am particularly interested in how safety standards and certification processes for robotic haircutting systems may evolve in the future. I am also curious whether augmented reality or virtual hairstyle preview systems could become part of the customer experience before the haircut is executed.

Thank you again for the opportunity. I look forward to learning more about this field.

Best regards,

Souptik Chattaraj